Where every number and every step comes from

Every figure and every design decision in this project falls into one of three boxes. The third box is the honest one.

Box	Meaning
Published	a named source states it
Convention	widely used in industry, no derivation behind it
Chosen	Claude picked it; nothing supports the exact value

Nothing here is settled. Items in the third box are candidates for removal or for grounding, on Rafael's order.

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1. The loop itself

The six steps are not one methodology. They are two established patterns joined together, and the joins are mine.

Step	Pattern	Box
flag, decide, label	the investigator feedback loop of card-fraud detection	Published
watch	performance and input monitoring from credit-risk practice	Convention
retrain, promote	champion and challenger from credit scoring	Convention

Flag, decide, label. In card fraud, a model ranks transactions, a fixed number go to human investigators, and their verdicts come back as labels for the next training round. This is the setting Carcillo, Le Borgne, Caelen and Bontempi describe and measure in Streaming active learning strategies for real-life credit card fraud detection. They score with Top-100 precision, for the same reason this project scores the weekly top five: the review budget, not accuracy, is what a desk actually has.

The companion handbook, Reproducible Machine Learning for Credit Card Fraud Detection (Le Borgne, Siblini, Lebichot and Bontempi, ULB, 2022), is the source for validating forward through time and for labels that arrive late.

Champion and challenger. The incumbent model keeps serving while a candidate is measured against it, and a winner is promoted. Described by Oracle, IBM and SAS.

Two caveats, both against this project. Those are vendor documents, not regulatory ones; a formal account of model governance would start from supervisory guidance such as the Federal Reserve's SR 11-7, which this project has not read. And in the usual champion-challenger setup a share of live traffic goes to the challenger. Here nothing is split: both models score the same weeks offline. That is a weaker test and the documents do not say it is acceptable.

What no source covers. The order of the six steps, and that drift alone may trigger a retraining. Those are mine.

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2. The error rates the simulation plants

These come from project 01, which plants the errors this project hunts.

Claim	Figure	Box	Source
duplicate or erroneous payments	1.5% of disbursements	Published, second-hand	APQC, measure 105905, as quoted by Corpay and CFO.com
the same, best and worst quartile	0.8% and 2.0%	Published, second-hand	Corpay
freight invoices carrying an error	5% to 10%	Claimed in the plan	Tompkins Ventures, **not verified in this pass**

Two problems with the first row. Nobody has read APQC's own page; every figure above is quoted by a vendor. And APQC's number is a share of *disbursements*, while this project applies it per *invoice*. Those are not the same denominator.

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3. The numbers that govern the loop

Name	Value	Box	What it rests on
`K`, invoices ranked first each week	5	Chosen	about as many as the simulation plants per week, so a perfect model scores 1
`DESK`, invoices checked by hand	25	Chosen	nothing
`LINE`, the drift threshold	0.25	Convention	the usual band: under 0.1 stable, 0.1 to 0.25 minor, over 0.25 a real shift (R `scorecard`)
`RUN`, weeks under the floor	3	Chosen	nothing
`CHALLENGE_WEEKS`	4	Chosen	nothing
`DRIFT_WEEKS`	8	Chosen	nothing
the floor	measured	Published method	the tenth percentile of the champion's own weekly precision

On 0.25. It is a rule of thumb, not a derived threshold, and it is criticised in the literature. Potgieter, Fourie and colleagues show the constant ignores sample size: it fires needlessly on small samples and stays quiet on real shifts in large ones, and they propose a chi-squared alternative instead (arXiv 2307.11878). This project uses the rule of thumb anyway, with a week against eight weeks, and says so here rather than pretending the number is principled.

On 5, 25, 3, 4 and 8. None of these is measured or sourced. Each was chosen because it sounded like a working week. The floor is the one number in this table that is derived from data.

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4. Claims that must be removed or grounded

Tracked in removals.md.

1. "Two people" at the desk. No headcount was decided or measured.
2. "5,600 invoices a year, about 200 wrong." The measured figure on one seed is 5,330 and 191, and no one counted the other four seeds.

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5. What this project does not claim

- It is not validated against any real accounts-payable data. Every invoice is invented by project 01.
- The analyst is played by the answer key, so the feedback loop is measured against a perfect reviewer who never errs and never tires.
- No supervisory or audit standard has been read, and none is met.

Bibliography

- Carcillo, F., Le Borgne, Y.-A., Caelen, O., Bontempi, G. *Streaming active learning strategies for real-life credit card fraud detection: assessment and visualization.* arXiv:1804.07481. https://arxiv.org/pdf/1804.07481
- Le Borgne, Y.-A., Siblini, W., Lebichot, B., Bontempi, G. *Reproducible Machine Learning for Credit Card Fraud Detection: a Practical Handbook.* Université Libre de Bruxelles, 2022. https://github.com/Fraud-Detection-Handbook/fraud-detection-handbook
- Potgieter, C. et al. *A statistical approach to population stability measurement.* arXiv:2307.11878. https://arxiv.org/pdf/2307.11878v4
- APQC, Open Standards Benchmarking, measure 105905, duplicate or erroneous payments, as quoted by Corpay and CFO.com. The primary page has not been read.
- Oracle Financial Services, *Champion Challenger.* https://docs.oracle.com/en/industries/financial-services/ofs-analytical-applications/credit-risk-analytics/81210/emoug/champion-challenger.html
- SAS, *Champion and challenger models in Enterprise Decision Manager.* https://support.sas.com/documentation/cdl/en/edmug/67015/HTML/default/p060qw8wi2u6mfn18ca2bvpl5hz8.htm
- IBM, *Champion challenger overview.* https://ibm.com/docs/en/SS69YH_8.2.1/cads_manager_ddita/model_management/thick/champion_challenger.html

Not yet read, and needed

- APQC's own benchmark page, for the duplicate-payment figure and its denominator.
- Tompkins Ventures, for the freight invoice error rate.
- Siddiqi, N. *Credit Risk Scorecards*, the PSI section, for where the 0.25 band actually comes from.
- Federal Reserve SR 11-7, *Guidance on Model Risk Management*, for what promoting a model is supposed to require.